

***MIMIC-IV* Dataset**

QA Benchmark for Coverage

Diversity in RAG Retrieval

Outline

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Why Build a Benchmark on medical data?

- ▶ Need to show coverage wins when candidate pools have multi-cluster structure
- ▶ Medical data is multi-faceted
 - Aspects: diagnosis, treatment, demographics, comorbidities
- ▶ Example of queries:
 - *"How is pneumonia managed in elderly patients with CKD (Complex Kidney Disease)?"*
 - *"What are the symptoms of stroke, and how does the treatment differ if the patient has a prior history of myocardial infarction?"*

Ideal Dataset for Diversity

Given a query, the ideal document pool candidates should have:

- ▶ **Redundant central cluster:** dominant topic, many similar documents
- ▶ **Complementary cluster(s):** other **relevant** aspects for the query
- ▶ **Niche cluster:** smaller set of **relevant** data points
- ▶ **Outliers:** noise, or genuine isolated points relevant to the query

Coverage: Facility-Location Objective

- ▶ Objective function

$$f(S) = \lambda \sum_{i \in S} utility(i) + (1 - \lambda) \sum_{i \in D} \max_{j \in S} sim(i, j)$$

- Every item in the candidate pool D has a representative in S
- Respects data distribution rather than maximizing spread between items

- ▶ Theoretical guarantees

- $g(S)$ is monotone submodular
- Greedy selection gives $(1 - 1/e) \approx 0.632$ approximation w.r.t. optimal solution
- No such guarantee exists for *dispersion* (NP-hard)

MIMIC-IV Dataset Overview

- ▶ *MIMIC-IV*: tabular clinical data (diagnoses, procedures, meds, labs, measurements)
- ▶ *MIMIC-IV-Note*: free-text data
 - **Discharge Summaries** (~330K): in-depth notes which overview a patient's history and course throughout a hospitalization
 - *Radiology Reports* (~2.3M): reports spanning x-ray, CT, MRI, and ultrasound
- ▶ Discharge summaries allow for design of multi-aspect queries requiring synthesis across patients
- ▶ Structured info in MIMIC-IV used for document pool construction and patient usable data
- ▶ De-identification: all PHI removed/replaced with ___ placeholders

Ideal Dataset for Coverage: MIMIC-IV?

Cluster Type	Possible MIMIC-IV candidates
Redundant central cluster	Common conditions → hundreds of similar notes
Complementary cluster(s)	Comorbidity combos (pneumonia + CKD + diabetes) → distinct management sub-groups
Niche cluster	Rare complications → few notes, unique reasoning
Outliers	ICD mismatches, boilerplate copies, noisy de-id artifacts

MIMIC-IV-Note Discharge Summaries

Section	Content	QA Value
Brief Hospital Course	Clinical reasoning per problem	High
History of Present Illness	Symptom narrative, temporal progression	High
Pertinent Results	Labs, imaging, micro interpretations	High
Discharge Diagnosis	Diagnostic labels (linkable to ICD)	High
Chief Complaint	1–2 lines, reason for admission	High (as embedding metadata)
Past Medical History	Comorbidities list	Medium
- Surgical / Invasive Procedure - Social / Family History - Discharge Medications - Physical Exams	Other minor pieces of information	Low

MIMIC-IV Structured Data

hosp module

Table	Content	Our use
diagnoses_icd	ICD-9/10 codes per admission , with seq_num priority	ICD labels + query scaffolding
d_icd_diagnoses	ICD code definitions (text descriptions)	Condition names for query templates
patients + admissions	Gender, age, date of death, admission type, insurance, race, discharge location	Patient data for additional context

Discharge summaries: what data to retain?

- ▶ **Brief Hospital Course** has highest QA value for complex medicine admissions with # sub-problem lists
 - Complex cases: management reasoning, drug choices with justifications, differentials
- ▶ Surgical/simple cases: formulaic procedural timelines, nearly content-free
 - These are noise in any candidate pool: no clinical reasoning
- ▶ **History of Present Illnesses** complements BHC: presentation context, symptom timeline, risk factors that BHC presupposes
- ▶ **Pertinent Results**: mostly raw lab noise, but imaging interpretations can be high-value

Experimental Pipeline

corpus construction → embeddings → query generation → evaluation

Phase 1: Corpus Construction

1. **Select conditions**

Query `diagnoses_icd` for ICD groups with $\geq N$ admissions

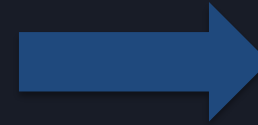
2. **Parse and chunk documents**

Split Discharge Summaries by section header

3. **Attach metadata**

Per chunk: section name, all ICD codes, demographics, admission type and details

4. **Deduplicate**



Output

**Per-condition
candidate pool D**

Phase 2: Embeddings Generation

- ▶ Raw chunks are too patient-specific for condition-level queries
- ▶ **Add context before chunks to capture full semantics**
- ▶ MIMIC metadata is structured → deterministic template to build context

Chunk example

```
Elderly male, White, admitted for Sepsis (SOFA 14).  
Chief complaint: fever, hypotension.  
Comorbidities: CHF, CKD, COPD.  
Section: Brief Hospital Course  
<section_content>
```

Phase 3: Query Generation via LLM

- ▶ Pick primary condition + 1–2 **modifier axes** to ensure multi-aspect
 - Comorbidity: co-occurring ICD codes (e.g. stroke + diabetes)
 - Demographic: age group, gender, race, etc.
 - Complication: post-surgical, recurrent, or secondary diagnoses
- ▶ LLM prompt: generate clinical management question requiring modifier knowledge
- ▶ **Pre-filter**: only keep queries where *coverage* $\neq$ *top-k*
 - assess overall suitability of coverage approaches for this dataset with this embedding strategy
 - $Jaccard(S_{cov}, S_{top-k}) \approx 1 \rightarrow \text{skip}$
 - $g(S_{cov}) - g(S_{top-k}) < \text{threshold} \rightarrow \text{skip}$
- ▶ **Gold answer generation**: LLM cites source chunks, each tagged with facet label

```
[
  "In patients with Type 2 diabetes and advanced kidney disease, how do clinicians manage the transition
  between pharmacological management of glycemic control and the initiation of renal replacement therapy, such as
  dialysis, and what specific lab values or symptoms typically trigger these changes?",

  {
    "fact": "Clinicians discontinue or hold oral hypoglycemics like metformin and glipizide in the setting of
    renal failure or decreased renal clearance to avoid complications such as lactic acidosis or hypoglycemia.",
    "facet_label": "pharmacological_transition_renal_failure",
    "chunk_ids": [ 50962, 101542, 32657... ]
  },
  {
    "fact": "Insulin doses are downtitrated or reduced when renal failure is present due to decreased renal
    clearance of insulin, which increases the risk of hypoglycemia.",
    "facet_label": "insulin_dose_adjustment_renal_failure",
    "chunk_ids": [ ... ]
  },
  {
    "fact": "Some patients completely cease the use of insulin or other diabetes medications upon the
    initiation of dialysis.",
    "facet_label": "dialysis_impact_on_glycemic_management",
    "chunk_ids": [ ... ]
  },
  {
    "fact": "Specific triggers for adjusting diabetes medications include severe hypoglycemia (e.g., BG 27-
    40), acute kidney injury (elevated creatinine), and the initiation of renal replacement therapy.",
    "facet_label": "triggers_for_medication_change",
    "chunk_ids": [ ... ]
  }
]
```

Phase 4: Evaluation

Methods: top-k (baseline), MMR, gMMR, facility-location

Metric	Definition	Measures
Aspect Recall	$AR(S) = \frac{ \{f \in F_{query}: \exists doc \in S \text{ with facet } f\} }{ F_{query} }$	Fraction of annotated facets covered by selected k chunks
Weighted Aspect Recall	retrieved gold chunks / total gold chunks for a facet	measures how completely each facet is covered
Fac-Loc Score	$g(S) = \sum_{i \in D} \max_{j \in S} sim(i, j)$	Representativeness of S over candidate pool D
Jaccard	$J(S_{cov}, S_{top-k})$	Overlap with baseline. $J \approx 1$: no change
GoldRecall	Gold doc fraction recovered	How many gold documents/facts are in the selection

Pros & Cons

Pros

- ❑ Multi-aspect queries tailored to coverage hypothesis
- ❑ Dataset seems suitable for coverage
- ❑ Gold sets with facet labels enable granular evaluation metrics

Cons

- ❑ Discharge notes may be complex to parse and interpret
- ❑ De-identification detrimental for semantics
- ❑ No guarantee about embedding distribution with the contextual prefix
- ❑ Query is still LLM-generated
- ❑ LLM gold annotation is expensive and noisy
- ❑ Facet labels depend on LLM judgment
- ❑ Local inference (DUA). Slower, smaller models
- ❑ Many moving parts before first result
- ❑ No existing benchmark to compare against
- ❑ human validation?

End

Extras

Pipeline Overview

Phase	Name	Key Actions	Output
1	Corpus Construction	Select ICD conditions, chunk notes by section, attach metadata, deduplicate	Per-condition candidate pools with metadata
2	Embeddings Generation	Prepend metadata prefix before embedding	Contextualized chunk vectors
3	Query Generation via LLM	LLM generates multi-aspect clinical questions; gold answers with facet labels	QA pairs with gold chunk sets
4	Evaluation	Aspect recall, Fac score, Jaccard, DocRec/FactRec	Coverage vs. dispersion comparison

Key Parameters

Parameter	Meaning	Notes
λ	Relevance–diversity tradeoff	$\lambda=1$: pure top-k, $\lambda=0$: pure coverage
N	Candidate pool size for re-ranking	Too small: no room to diversify; too large: noise enters
k	Number of chunks to select	Final selection size
T_{max}	Token budget	Constrains total context length for LLM
chunk_tokens / stride	Chunk size and overlap	Controls granularity of information units

Gold Response Generation via LLMs

Map

- ▶ Split candidate pool into batches of ~50 chunks (~10K tokens)
- ▶ LLM extracts facts relevant to the query from each batch
- ▶ Each fact is cited back to its source chunk(s)

Reduce

- ▶ Merge pass synthesizes across all batches
- ▶ Deduplicate facets, resolve conflicting citations
- ▶ Final gold answer with citations + facet labels per chunk

Facet labels: Each gold chunk tagged with aspect (e.g., `general_symptoms`, `cardiac_confounding`, `treatment_difference`) → enables aspect recall metric

Quality verifier: Separate reader-LLM attempts top-k answer; if it succeeds → query too easy → discard

Computational Feasibility

- ▶ Full corpus: $\sim 3\text{M}$ chunks $\rightarrow O(|D| \cdot k^2)$ facility-location is impractical
- ▶ Two-stage pipeline: HNSW retrieves $N \gg k$ candidates, then re-rank
- ▶ Re-ranking on $N=100\text{--}1000$ candidates is fast
- ▶ N is a tunable parameter alongside λ and k

- ▶ **Alternative:** build sub-corpus with limited scope, adding also distractors (like HotpotQA and 2WikiMultihopQA)

Cluster Validation Approaches

- ▶ UMAP + HDBSCAN pipeline on embeddings
- ▶ Validate with DBCV, cluster size/density ratios, noise fraction
- ▶ Cross-reference clusters with ICD sub-codes for semantic meaning

- ▶ Simpler alternative: Jaccard divergence + facility-location gap + aspect recall
- ▶ Full clustering may not be necessary - downstream metrics are the real validation

Phase 2: Embedding models for medical domain

Model	Dim	Context
BioLORD-2023	768	512
MedEmbed (large)	1024	512
BioClinical ModernBERT	768	8,192